

Angitha Mriduraj

Toronto, Canada
a.mriduraj@mail.utoronto.ca

EDUCATION

- University of Toronto** September 2024– Present
PhD Student in Fundamental Immunology
- University of Toronto** September 2019 – 2024
Honors BSc in Cell and Molecular Biology (focus in Stem Cells and Developmental Biology).

RESEARCH EXPERIENCE

- Research Student (Independent Project)** | *The Hospital for Sick Children* - Grunebaum Lab May 2023 – Present
Investigated if there were changes in the hematopoietic potential of lineage-negative HSCs lacking purine nucleoside phosphorylase (PNP) prior to the administration of lentiviral gene therapy.
- Performed colony forming unit assays
 - Conducted vector copy number analysis, qPCR and Lowry protein assays
 - Performed lineage depletion and cultured mesenchymal stem cells
 - Collected bone marrow from mice
- Research Student (Independent Project)** | *University of Toronto* - Bruce Lab September 2022 – April 2023
Worked with zebrafish embryos to examine the spatiotemporal expression pattern of *camsap* and other patterning genes to understand the function of microtubule binding protein Camsap2.
- Performed whole mount *in situ* hybridization
 - Generated DNA constructs and *in situ* hybridization probes
 - Utilized confocal microscopy for live imaging
 - Genotyped and phenotyped mutant embryos
 - Experienced in DNA extraction, PCR and restriction digests
- Research Student (Independent Project)** | *University of Toronto* - Ehrhardt Lab September 2021 – July 2022
Investigated novel cell surface receptors recognized by Variable Lymphocyte Receptors (VLRs) from Sea Lamprey and analyzed their affinity for human cytokines.
- Performed transformations, transfections, and electroporation
 - Worked with HEK293T and B-JAB B-cell lines
 - Conducted western blot analysis, DNA gel electrophoreses and extraction
 - Gained exposure to PCR and affinity chromatography
 - Experienced in tissue culture and tissue staining
 - Utilized yeast surface display
- Research Assistant** | *University of Toronto* - Gilbert Lab May 2021 – August 2021
Quantified various features of muscle cell cultures stained by immunofluorescence to identify muscle cell shape, size, and protein expression.
- Familiar with Fiji/Image J and Microsoft Excel
 - Perform statistical analysis
- Research Assistant** | *University of British Columbia* - Jakobi Lab May 2021 - February 2022
Participated in a scooping literature review for publications discussing the effects of cannabis/cannabinoids on synaptic integration of motor commands in the spinal cord.
- Evaluated and screened publications against inclusion criteria
 - Used Covidence and Microsoft Excel
- Research Assistant** | *University of British Columbia* - Donkelaar Lab October 2020 - April 2021
Assisted in coding 221 survey results according to the Theoretical Domains Framework (TDF) to identify barriers which

prevent discussions about traumatic brain injuries by staff at women's shelters with clients.

- Communicated and justified my results in meetings.

PUBLICATIONS

- Nicol B, Adhikari SP, Shwed A, Ashton S, **Mriduraj A**, Mason K, Gainforth HL, Babul S, van Donkelaar P. 2023. The concussion awareness training tool for women's support workers improves knowledge of intimate partner violence-caused Brain Injury. *INQUIRY: The Journal of Health Care Organization, Provision, and Financing*.
- Haynes EM, Kalmar JM, Vis-Dunbar M, Crosby KM, **Mriduraj A**, Jakobi JM. 2023. A systematic review of how cannabinoids affect motoneuron output. *Journal of Neurophysiology*.

POSTERS & PRESENTATIONS

- **Mriduraj A**, Ehrhardt G. (2022). Elucidating the novel cell surface antigens recognized by variable lymphocyte receptors. Undergraduate Thesis Symposium. Toronto, Canada. PowerPoint presentation.
- **Mriduraj A**, Ehrhardt G. (2022). Isolating TNF reactive Variable Lymphocyte Receptors (VLRs) of the evolutionarily distant sea lamprey using Yeast Surface Display. Summer Research Symposium. Toronto, Canada. PowerPoint presentation.
- **Mriduraj A**, Wan H, Bruce A. (2023). Characterizing the impact of *camsap2a* knockout on patterning during zebrafish epiboly. Undergraduate Thesis Symposium. Toronto, Canada. Poster presentation.
*Received F Michael Barrett outstanding poster award
- **Mriduraj A**, Loves R, Grunebaum E. (2023) Characterizing clonogenic potential, enzyme activity, and protein production in a Purine Nucleoside Phosphorylase (PNP) deficient mouse model. Sickkids Summer Research Symposium. Toronto, Canada. Poster presentation.
- Bruce A, **Mriduraj A**, Wan H. (2023). The role of Camsap2 in the yolk cell during zebrafish gastrulation. 82nd Annual Society for Developmental Biology Conference. Chicago, USA. Poster presentation.
*Not the presenter

WORK EXPERIENCE

English and Math teacher | *Marilake Academy* September 2020 - June 2021 & September 2022 – Present
During the academic year, I dedicate my Saturdays to teaching English (grades 4/5/6/7/9/10/11/12) and Math (grades 4/7), accumulating a total of 174 students taught over 620 hours (270 hours paid and 350 unpaid).

- Designed an engaging curriculum
- Planned and conducted captivating presentations
- Evaluated worksheets weekly and provided tailored feedback
- Maintained proactive communication with parents and staff outside of regular class hours

COMMUNITY SERVICE

Content Team | *IMMspire* September 2024 - Present
Made the theory behind lab techniques more accessible to high school students by preparing a workshop for the event day.

- Collaborated with peers to create engaging and informative content
- Contributed to brainstorming sessions to develop content and activities

Mentorship Program Coordinator | *Graduate Peer Support Network* September 2024 - Present
I work with another peer to recruit graduate student mentees and mentors.

- Ensure the mentors have proper training
 - Meet with the rest of the team monthly
- Host socials

Volunteer Tutor | *Parsec Youth Network* August 2020 - Present

Provided 165 hours of dedicated tutoring support to an elementary student from 5th grade to 7th grade.

- Independently tailored lessons and assignments to target specific weaknesses
- Fostered a strong connection with the student and learned about the student's background and family
- Continuously adapted my teaching style to their evolving growth and unique requirements.

Volunteer | *Fort York Food Bank*

May 2022 – April 2024

Devoted 200 hours to volunteering at one of Toronto's busiest food banks.

- Engaged with clients while adhering to food distribution guidelines
- Maintained composure and displayed empathy with agitated clients.
- Tailored client assistance by asking about food preferences.
- Actively listened and utilized resources to overcome language barriers.
- Thrived in busy sections while also supporting new volunteers.

Highschool Mentorship Program Director | *Women in Science and Engineering*

September 2022 – May 2024

I oversee and manage the high school mentorship program, encompassing the recruitment of high school student mentees and the planning of a minimum of two events per semester.

- Recruited and paired 17 mentors and 33 mentees

Vice President | *UofTress*

September 2021 – May 2024

Dedicated 55 hours to UofTrees, a sustainability-focused club, actively promoting environmental sustainability.

- Transitioned from Communications Director to Vice President of the club.
- Led a team of 15 members to enhance the club's social media presence and outreach.
- Collaborated with the Marketing VP to post engaging weekly infographics on Instagram
- Increased following to 1151
- Organized and executed 10 events (talks, workshops and challenges) with the team.

Mentor | *University Collage E-mentor*

May 2020 - September 2023

Mentored a total of 223 incoming first-year university students in a program designed to support and guide during their transition to university life, spanning from May to September each year.

- Established a Discord server
- Fostered engagement from students worldwide
- Discussed concerns about university life
- Delivered guidance on course selection and study techniques
- Promoted extra resources in biweekly newsletters and meetings
- Offered career advice and feedback on resumes, cover letters and interview preparation.

Event Coordinator and Member | *Stem Cells and Developmental Biology Focus Group*

September 2020 - Present

As a member of the focus group, I took the initiative to plan a Q&A session for aspiring graduate students, exercising full autonomy in both execution and event planning (10 hours total).

- Securing 8 graduate student panelists from various labs
- Led a 3-person team to efficiently coordinate and promote the event
- Provided prompts and hosted the session for 18 participants.
- Collaborated with a faculty member for guidance

Mentor | *Cells and Systems Biology Student Union Mentorship Program*

September 2021 - April 2023

Dedicated 20 hours to mentoring a total of four students within my department.

- Checked in bi-weekly
- Provided academic guidance, tailored to each student's unique learning strategies and circumstances
- Addressed personal concerns and connected them to additional resources.
- Enhanced mentoring skills in a 5-hour optional seminar series.

- Proactively actively engaged with inactive mentees.

Mentor | *Women in Science and Engineering*

September 2020 - April 2023

Collaborated with a peer to organize and facilitate bi-weekly meetings for 10 mentees, contributing a total of 25 hours.

- Nurtured relationships to inspire and empower mentees
- Held informative sessions on personal growth, opportunities, and career paths
- Attended optional seminars to better my ability to support and guide mentees

Senior Volunteer | *Vernon Jubilee Hospital Auxiliary Youth Healthcare Volunteer Program*

September 2014- June 2019

Volunteered a total of 133 hours over four years, dedicating time once a week at my local hospital.

- Assisted in training and task allocation for 7 new volunteers
- Enhanced the patient experience by providing reading materials, refreshments and warm blankets
- Offered general support for staff & ensured residents were comfortable during mealtime rushes
- Organized card games & outdoor breaks for residents.